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Ugonin P inhibits lung cancer motility by suppressing DPP-4 expression via promoting the synthesis of miR-130b-5p

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ABSTRACT

Lung cancer is the leading cause of cancer-related death worldwide, and the survival rate of metastatic lung cancer is exceedingly low. *Helminthostatchys Zeylanica* (*H. Zeylanica*) is a Chinese herbal medicine renowned for its anti-inflammatory, immunomodulatory, and anti-cancer activities in various cellular and animal studies. The current study evaluated the effects of *H. Zeylanica* derivatives on lung cancer cells. We determined that dipeptidyl peptidase-4 (DPP-4) expression levels were higher in lung cancer tissues than in normal tissues. We also determined that DPP-4 expression levels were higher in the metastatic stage and strongly correlated with lung cancer survival rates. An *H. Zeylanica* derivative (ugonin P) was shown to inhibit DPP-4 mRNA and protein expression in two lung cancer cell lines in a dose-dependent manner. Ugonin P was shown to decrease migration and invasion activities in lung cancer cells while promoting the synthesis of miR-130b-5p, which was found to negatively regulate DPP-4 protein expression and cell motility in lung cancer. We determined that ugonin P suppresses the DPP-4-dependent migration and invasion of lung cancer cells by downregulating the RAF/MEK/ERK signalling pathway and enhancing the expression of miR-130b-5p. This study provides compelling evidence that ugonin P could be used to develop novel therapeutic agents for the treatment of lung cancer.

1. Introduction

Lung cancer is the leading cause of cancer-related death worldwide with roughly 2 million diagnoses and 1.8 million deaths each year [1]. Lung cancer is a complex illness with a variety of subtypes [2], all of which have been linked to smoking. Adenocarcinoma is the most prevalent among non-smokers [3]. Based on histology, lung cancers are classified as small-cell lung cancer (SCLC) or non-small cell lung cancer (NSCLC). NSCLC accounts for 83 % of cases, while SCLC accounts for 13 % [4]. Metastasis refers to the formation of secondary tumors distal to

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the original cancer site via migration through the lymphatic or cardiovascular system [5]. Metastasis via the lymphatic system is more gradual than that through the circulatory system [6]. Nearly 90 % of cancer deaths can be attributed to metastasis [7].

The development of molecular targeted therapies and immunotherapies over the last two decades has greatly improved the prognosis for lung cancer patients [8]. For lung cancer stages I or II, surgical removal is recommended. For stages II-III, adjuvant platinum-based chemotherapy has been shown to improve 5-year survival by 5.4 %; however, relapse and adverse side effects are common [9]. For late-stage lung cancer, powerful medications such as osimertinib [10], pembrolizumab [11], and lorlatinib [8,12] have largely been discontinued due to serious adverse effects.

Helminthostatchys Zeylanica (*H. Zeylanica*) is a Chinese herbal medicine renowned for its curative properties, including hepatoprotective [13], anti-inflammation [14], anti-oxidation, anti-osteoporosis [14], immunomodulatory, and neuroprotective effects [15]. One study reported that *H. Zeylanica* suppresses the production of pro-inflammatory cytokines stimulated by lipopolysaccharides in macrophage cells by targeting the NF-κB signaling pathway [16]. Another study reported that *H. Zeylanica* can mitigate liver damage induced by oxidative stress and/or inflammation [17]. *H. Zeylanica* has also been shown to induce apoptosis, autophagy, and cell cycle arrest in gastric cancer cells while hindering cell migration [18]. This is the first study to examine the effects of *H. zeylanica* derivatives in lung cancer. We determined that the *H. zeylanica* derivative ugonin P inhibits dipeptidyl peptidase-4 (DPP-4)-dependent motility in lung cancer cells, inversely regulates miR-130b-5p expression, and inhibits RAF/MEK/ERK signaling cascades. This paper provides compelling evidence that ugonin P could be used to develop novel therapeutic agents for the treatment of lung cancer.

2. Materials and methods

2.1. Materials

Matrigel was supplied by BD Biosciences (Bedford, MA, USA). A TRIzol kit was purchased from MDBio Inc. (Taipei, Taiwan). Transwell chambers with 8 μm pore sizes were purchased from Corning (Kennebunk, ME, USA). A Novolink Polymer Detection Systems kit was purchased from Leica Biosystems. Anti-mouse and anti-rabbit secondary antibodies were respectively obtained from Biolegend (#11982) and Santa Cruz Biotechnology Inc. (SC-2357; CA, USA). A Human Protease Antibody Array kit was supplied by RayBiotech, Inc. The MTT solution 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide, dimeth yl sulfoxide (DMSO), and other chemical reagents not already mentioned were supplied by Sigma-Aldrich (St Louis, MO, USA).

2.2. Cell culture

The human lung cancer cell lines A549 and CL1-5 were obtained from the American Type Culture Collection (Manassas, VA, USA) and cultured in Dulbecco's modified Eagle's medium (DMEM) (Gibco Life Technologies Corporation, Grand Island, NY, USA) supplemented with 10 % fetal bovine serum (FBS) (Lonza, Walkersville, MD, USA) and streptomycin/penicillin at 100 U/mL. Cell incubation was conducted in a humidified atmosphere of 37 °C under 5 % CO_2.

2.3. MTT assay

Lung cancer cells of human origin were treated with ugonin P at various concentrations (0.1, 0.3, 1, or 3 μM) for 24 h and then exposed to MTT solution for 2 h. DMSO was used to facilitate the detection using a microplate reader at a wavelength of 540 nm (BioTek, Winooski, VT, USA) [19].

2.4. Migration and invasion assays

Cells were treated with ugonin P at various concentrations (0.1, 0.3, 1, or 3 μM) or treated with pharmacological activators, including a RAF activator [(Coumermycin A1; Santa Cruz Biotechnology, Inc., CA, USA), MEK activator (PAFC-16), an ERK activator (Ceramide C6), or transfected with a miRNA inhibitor (hsa-miR-130b-5p; Thermo Fisher Scientific, Waltham, Massachusetts, USA)]. Invasion assays were performed using a transwell chamber after the cells had been incubated at 37 °C under 5 % CO_2 for 24 h [20]. The filter membranes in the transwell inserts were pre-coated with Matrigel to create a thin gel layer. The cells were then fixed in 3.7 % formaldehyde and stained with 0.5 % crystal violet solution (Merck, Darmstadt, Germany). Stained transwell inserts were examined under a visible-light microscope and photographed for use in quantifying migratory activity using ImageJ software (MacBiophotonics) [21–23].

2.5. Tissue arrays

Human lung cancer tissue array sections were deparaffinized using xylene and rehydrated using ethanol prior to immunohistochemical (IHC) staining. Deparaffinized sections were incubated with secondary antibodies (1:200) for 1 h at room temperature and then immunoassayed using DPP-4 antibodies (1:200) overnight. All tissues were stained using 3,3-diaminobenzidine and photographed using a light microscope [22,24,25]. IHC positive staining intensity was quantified using ImageJ software (MacBiophotonics).

2.6. Bioinformatic analysis

The GSE118370 dataset from the Gene Expression Omnibus (GEO) database was used to examine the expression of metastatic factors in clinical samples of lung cancer and normal cells. The GSE53882 dataset was used to examine miRNA expression profiles [26].

Records downloaded from the Kaplan-Meier database were used to examine the correlations between DPP-4 protein expression levels and survival prognoses. Records from the Gene Expression Profiling Interactive Analysis (https://gepia2.cancer-pku.cn/) database were used to examine DPP-4 expression levels in normal lung cells and adenocarcinoma cells [27].

2.7. Western blot and human protease arrays

Following the incubation of lung cancer cells with ugonin P for 24 h, total proteins were extracted using RIPA lysis buffer for quantification using the Thermo Scientific™ Pierce™ BCA Protein Assay Kit. After separating the resolved proteins using SDS-PAGE, the gels were transferred to polyvinylidene difluoride (PVDF) membranes. After blotting the membranes with skimmed milk (at 1:1000 dilution) for 1 h, the cells were incubated with primary antibodies overnight followed by incubation with secondary antibodies [21,28]. See Supplementary Table S1 for a list of antibodies used in this study.

Using a human protease protein array (obtained from Ray biotech), A549 cells were incubated with ugonin P (3 μM) for 24 h, after which total proteins were extracted using RIPA lysis buffer for quantification and subsequent incubation on blots in accordance with the manufacturer's protocols. Enhanced chemiluminescence images of the blots were then obtained using a UVP BioSpectrum Imaging System (UVP, Upland, CA, USA) [28,29]. The quantification of Western blot results was performed using ImageJ software [25].

2.8. mRNA and miRNA analysis

After incubating lung cancer cells with ugonin P for 24 h, total RNA was extracted using TRIzol reagent. Reverse transcription was performed using 1 μg of total RNA transcribed into complementary DNA

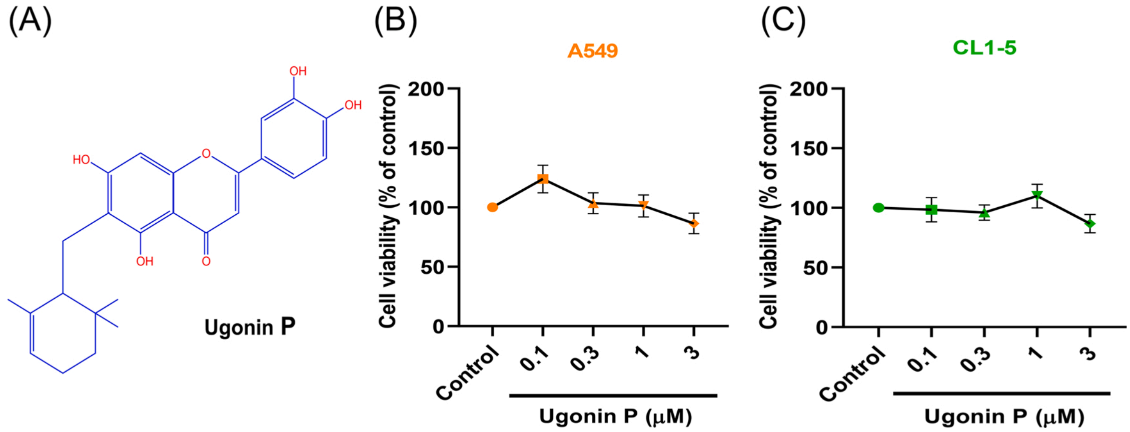

Fig. 1. Non-cytotoxic effects of ugonin P. (A) Chemical makeup of ugonin P. (B&C) A549 and CL1–5 lung cancer cell lines were treated with ugonin P (0.1, 0.3, 1, and 3 μM) for 24 h, the cell viability was examined by MTT assay (n = 3). Results are expressed as the means ± SD.

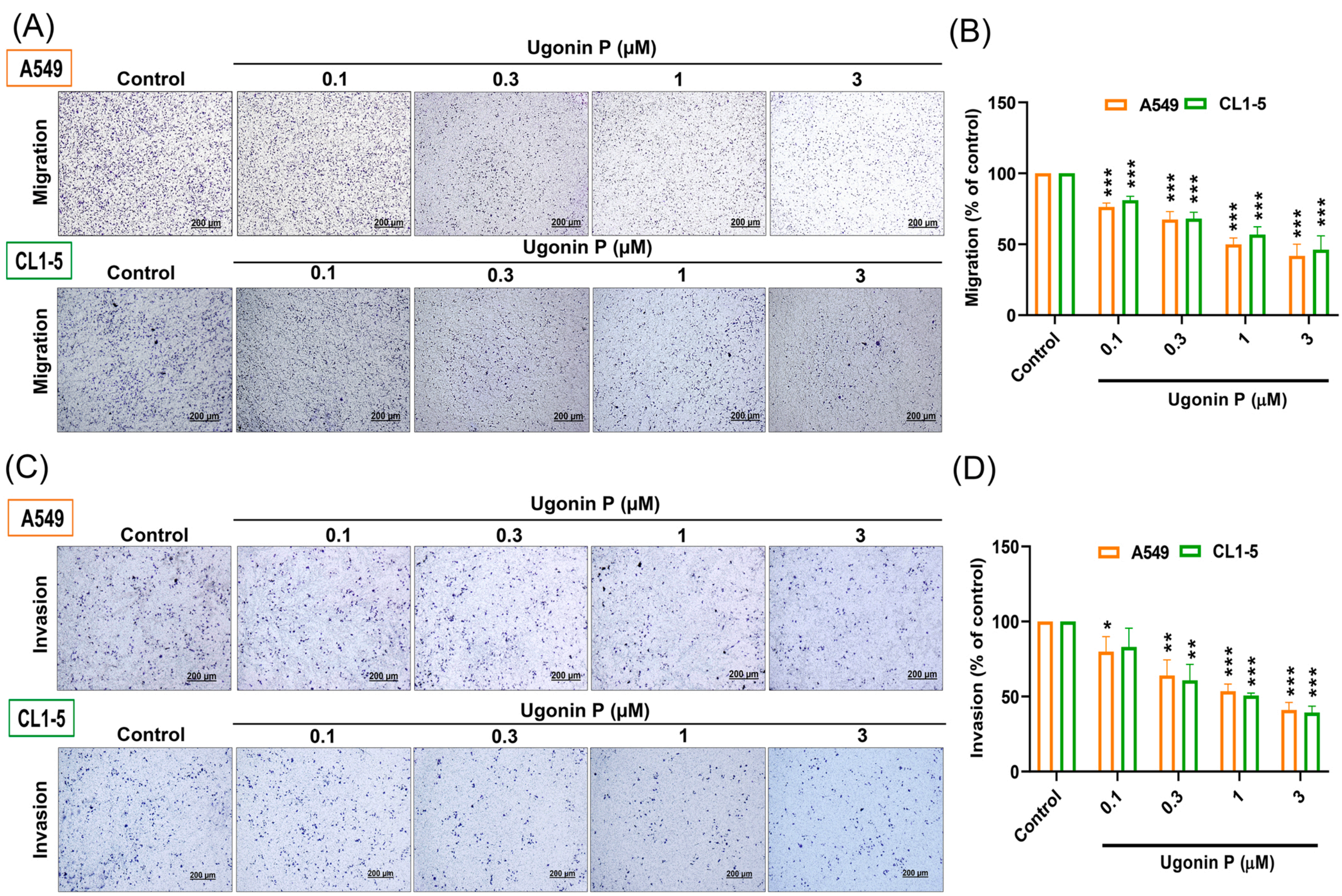

Fig. 2. Ugonin P inhibits lung cancer cell migration and invasion. Transwell assay results indicating in vitro migratory (A&B) and invasive capabilities (C&D) of A549 and CL1–5 cells after incubation with ugonin P at indicated concentrations for 24 h (n = 3). Cell migration was quantified by ImageJ software. Results are expressed as the means ± SD. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$ vs. control group.

using oligonucleotide primers and the Mir-X™ miRNA First-Strand Synthesis kit (Terra Bella Avenue; Mountain View, CA, USA). Quantitative real-time polymerase chain reaction (qPCR) was performed using the Taqman® One-Step QRT-PCR Master Mix (Applied Biosystems). qRT-PCR assays were performed using the StepOnePlus™ Real-Time PCR System (Applied Biosystems) [22,24]. Details pertaining to primer sequences and gene names are listed in Supplementary Table S2.

2.9. Statistical analysis

All values are presented as the mean ± standard deviation (S.D.). The Student's *t*-test was used to identify significant differences between groups. One-way analysis of variance (ANOVA) with post hoc Bonferroni correction for the comparison of multiple groups by using GraphPad Prism 8.2 (GraphPad Software, San Diego, CA, USA). Differences were considered significant if the *p*-value was < 0.05.

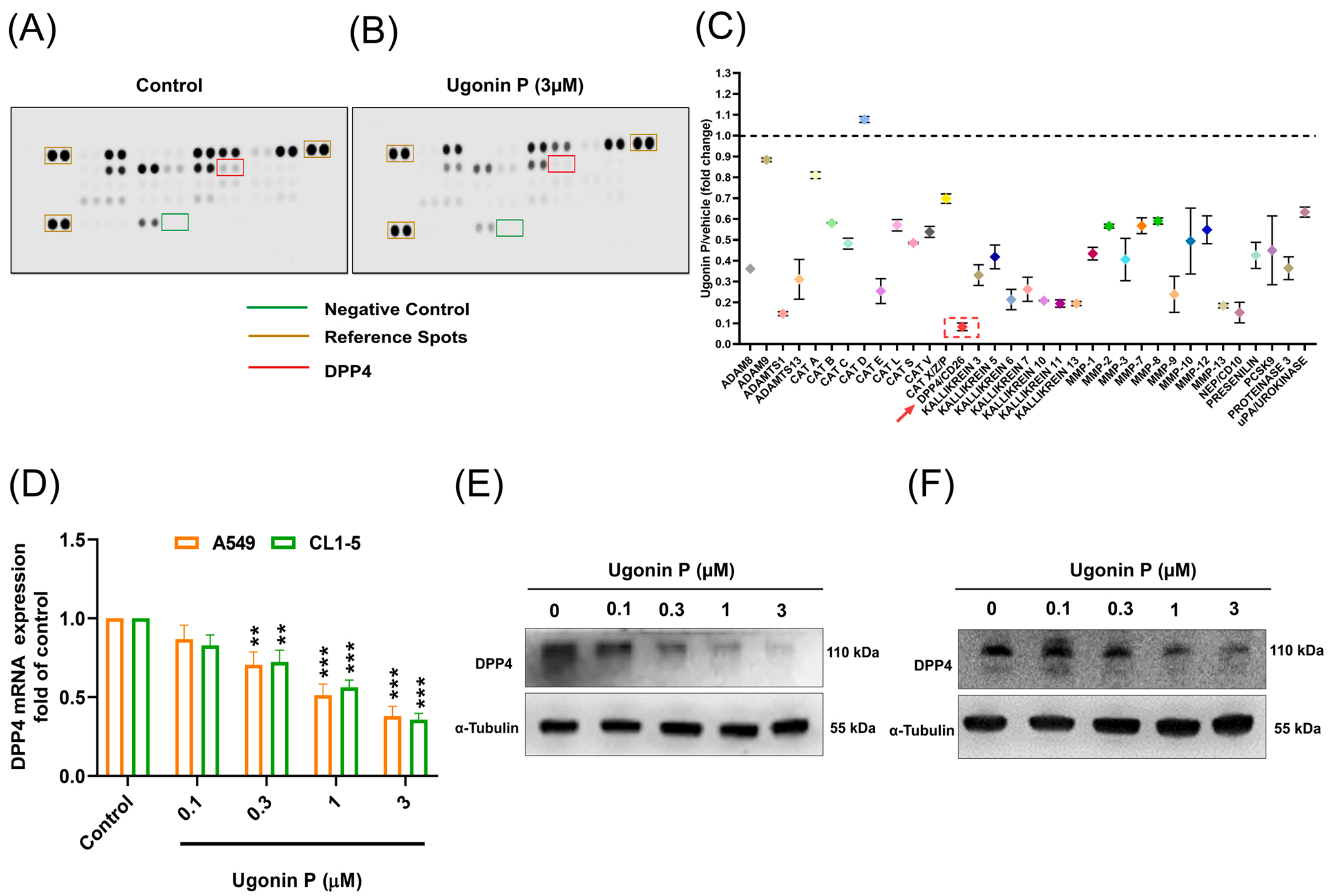

Fig. 3. Ugonin P inhibits DPP-4 expression. Results of human protease protein array indicating the expression of 35 protease factors in (A-C) A549 cells incubated with ugonin P (3 μM) for 24 h. 35 protease factors was quantified by ImageJ software. (D-F) A549 and CL1–5 cells incubated with ugonin P (3 μM) for 24 h. DPP-4 mRNA and protein expression were respectively evaluated via qPCR and Western blot analysis (n = 3). α-Tubulin was used as loading control. Results are expressed as the means ± SD. **p* < 0.05; ***p* < 0.01; ****p* < 0.001 vs. control group.

3. Results

3.1. Ugonin P inhibits the motility of lung cancer cells

Cancer metastasis is an important indicator of advanced malignancy and the main cause of therapeutic failure [7]. Metastasis is a multi-stage process involving cell migration, invasion, extravasation, and proliferation [30]. To determine the role of ugonin P in lung cancer metastasis, we began by studying the chemical makeup and structure of ugonin P (Fig. 1A). We then examined the viability of lung cancer cells (A549 and CL1-5) when exposed to ugonin P at various concentrations (0.1, 0.3, 1, or 3 μM) (Fig. 1B&C) and contrasted the results with those from untreated controls. Transwell assays were used to determine the inhibitory effects of ugonin P on lung cancer motility. Incubation of cancer cells with ugonin P for 24 h was shown to suppress cell migration in a dose-dependent manner (Fig. 2A&B). The in vitro invasion assay is a common method used to mimic cancer invasion and metastasis in vivo [31]. Ugonin P also inhibited cell invasive ability through Matrigel (Fig. 2C&D). In addition, ugonin P treatment diminished cell migration and invasion in MDA-MB-231 breast cancer cells (Supplementary Fig. S1). These findings suggest that ugonin P inhibits lung cancer cell motility without affecting viability.

A human protease protein array was used to identify molecular candidates involved in ugonin P-mediated anti-migratory activity. Incubation of A549 cells with ugonin P at 3 μM significantly lowered the expression of several protease proteins, particularly DPP-4 (Fig. 3A-C). qPCR and Western blot results provided further evidence that in both lung cancer cell lines, ugonin P lowered DPP-4 expression levels in a dose-dependent manner (Fig. 3D-F).

3.2. Ugonin P inhibits DPP-4-dependent migration and invasion by lung cancer cells

DPP-4 is a ubiquitously expressed membrane-bound glycoprotein with a diversity of biological functions. DPP-4 is expressed in many solid tumors and hematologic malignancies. It has also been identified as a potential biomarker of tumor progression [32]. Our use of bioinformatics tools to investigate the clinical relevance of DPP-4 in lung cancer revealed that DPP-4 expression levels were significantly higher in lung cancer cells than in normal lung cells (Fig. 4A). We also investigated the expression levels of DPP-4 in clinical samples. Staining a human lung cancer tissue array with DPP-4 antibodies revealed that DPP-4 expression levels were significantly higher in lung cancer cells than in normal control or adjacent tissue samples (Fig. 4B-C). Using the GEO database, we also determined that DPP-4 expression levels were higher in the metastasis stage than in pre-metastasis stages (Fig. 4D). Furthermore, we observed a significant negative correlation between DPP-4 expression levels and overall survival (Fig. 4E).

We hypothesized that DPP-4 is associated with cellular movement and the spread of cancer cells [33]. By transfecting lung cancer cells with DPP-4 plasmids, to investigate DPP-4 in ugonin P-inhibited lung cancer cells motility. As expected, DPP-4 overexpression (compared to the control group) was correlated with DPP-4 mRNA and protein expression (Fig. 5A-C). However, we also found that DPP-4 overexpression reversed

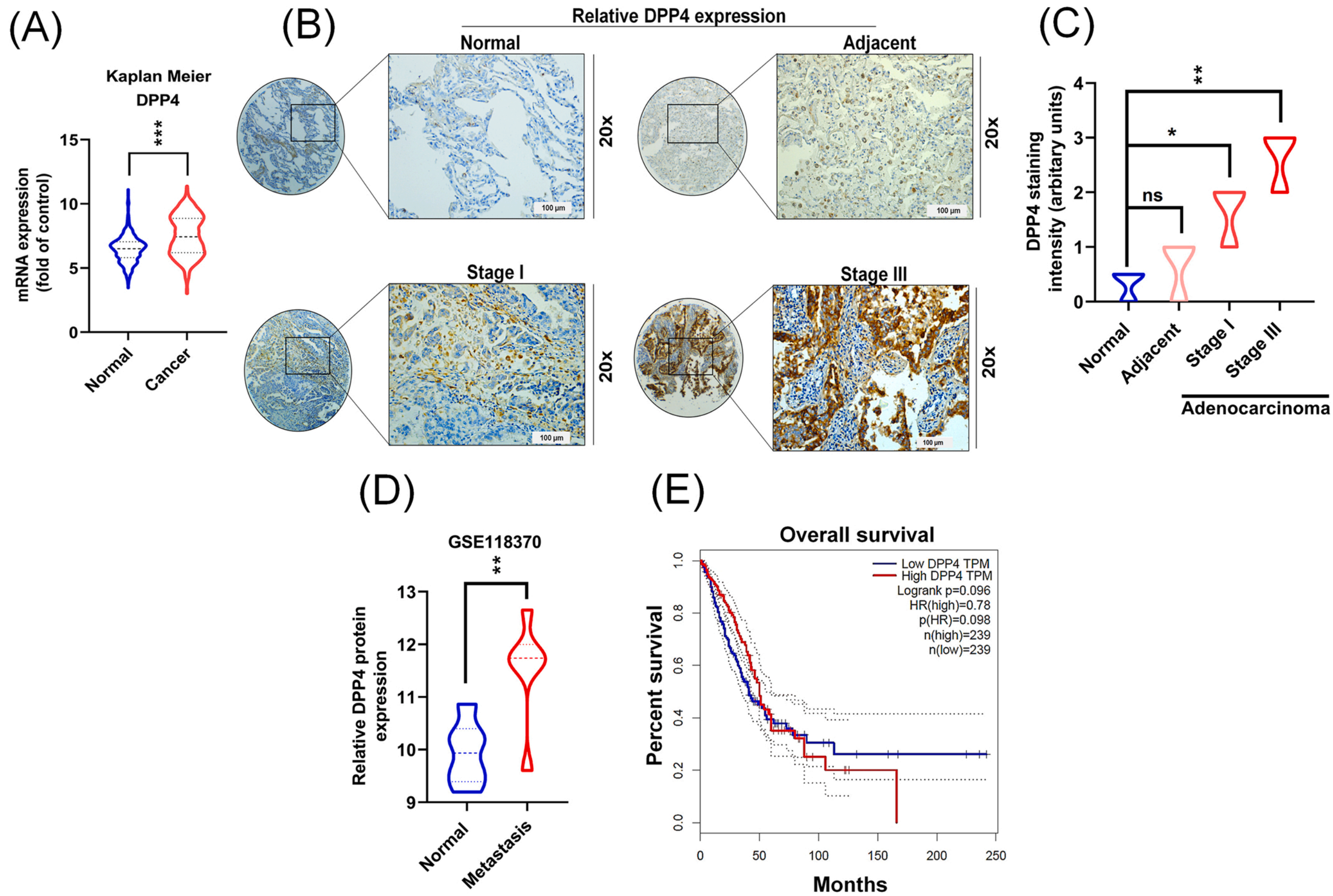

Fig. 4. Correlation between DPP-4 levels and lung cancer metastasis and survival. (A) DPP-4 expression levels in lung cancer samples and normal samples based on data from the Kaplan-Meier database. (B&C) Human tissue array was stained with DDP-4 antibody. DPP-4 staining results of human lung cancer tissue array quantified using ImageJ software. The DPP-4 levels from GEO and GEPIA dataset was analysed. (D) Comparison of DPP-4 expression in metastasis tissue samples and normal lung tissue based on the GEO database. (E) Overall survival as a function of DPP-4 expression in patients with lung cancer based on the GEPIA dataset. Results are expressed as the means ± SD. $*p < 0.05$; $**p < 0.01$; $***p < 0.001$ vs. normal group.

the inhibitory effects of ugonin P on DPP-4 expression, cell migration, and cell invasion (Fig. 5), thereby indicating that ugonin P inhibits DPP-4-dependent lung cancer motility.

3.3. Ugonin P inhibits lung cancer cell motility by downregulating the RAF/MEK/ERK signaling pathways

The RAF/MEK/ERK pathway plays a key role in the progression of lung cancer via cell proliferation, migration, invasion, and drug resistance [34,35]. We assessed the relative contributions of the respective signalling pathways in the suppression of DPP-4-dependent lung cancer migration and invasion by ugonin P. Western blot analysis revealed that ugonin P significantly reduced the phosphorylation of RAF, MEK, and ERK in a concentration-dependent manner (Fig. 6A-B). Subsequent analysis using pharmacological activators [(RAF; Coumermycin A1), (MEK; PAFC-16), and (ERK; Ceramide C6)] revealed a reversal of the ugonin P induced inhibition of DPP-4 mRNA and protein expression (Fig. 6C-D) as well as activities related to lung cancer migration and invasion (Fig. 6E-F). In addition, RAF, MEK and ERK siRNA all augmented ugonin P-inhibited DPP-4 expression and cell migration (Supplementary Fig. S2). Taken together, these results demonstrate that ugonin P inhibits the motility of lung cancer cells by downregulating the RAF/MEK/ERK signaling pathway.

3.4. Ugonin P inhibits lung cancer motility through the upregulation of miR-130b–5p

miRNAs have been broadly implicated in cancer development, metastasis, angiogenesis, and drug resistance [36]. Numerous studies have reported that miRNAs can act as tumor suppressors by inhibiting the expression of genes involved in promoting tumor growth, cell proliferation, angiogenesis, and metastasis [26]. In the current study, we investigated the role of miRNA in the ugonin P-mediated inhibition of lung cancer motility by comparing the expression levels of miRNAs in ugonin P-treated samples and untreated controls. This miRNA sequencing process identified several miRNAs showing significant differences in expression levels (Fig. 7A). We then ran three miRNA prediction programs (Target Scan, miRmap, and miRDP) using the collected miRNA sequencing data. We identified three candidate miRNAs that bind to the 3′UTRs of DPP-4, including miR-130b-5p, miR-5006–3p, and miR-758–3p (Fig. 7B). Ugonin P treatment was shown to promote miR-130b-5p expression in a concentration-dependent manner, compared to other two miRNAs (Fig. 7C-D). This prompted us to use GEO database records to evaluate the clinical relevance of miR-130b-5p. Overall, we determined that miR-130b-5p expression levels were lower in samples from lung cancer patients than in normal controls (Fig. 7E-F).

These findings prompted further research into the role of miR-130b-5p in ugonin P-inhibited lung cancer motility in the presence of a miRNA-130b-5p inhibitor. Cells transfected with the miR-130b-5p inhibitor were shown to antagonize the inhibitory effects of ugonin P on

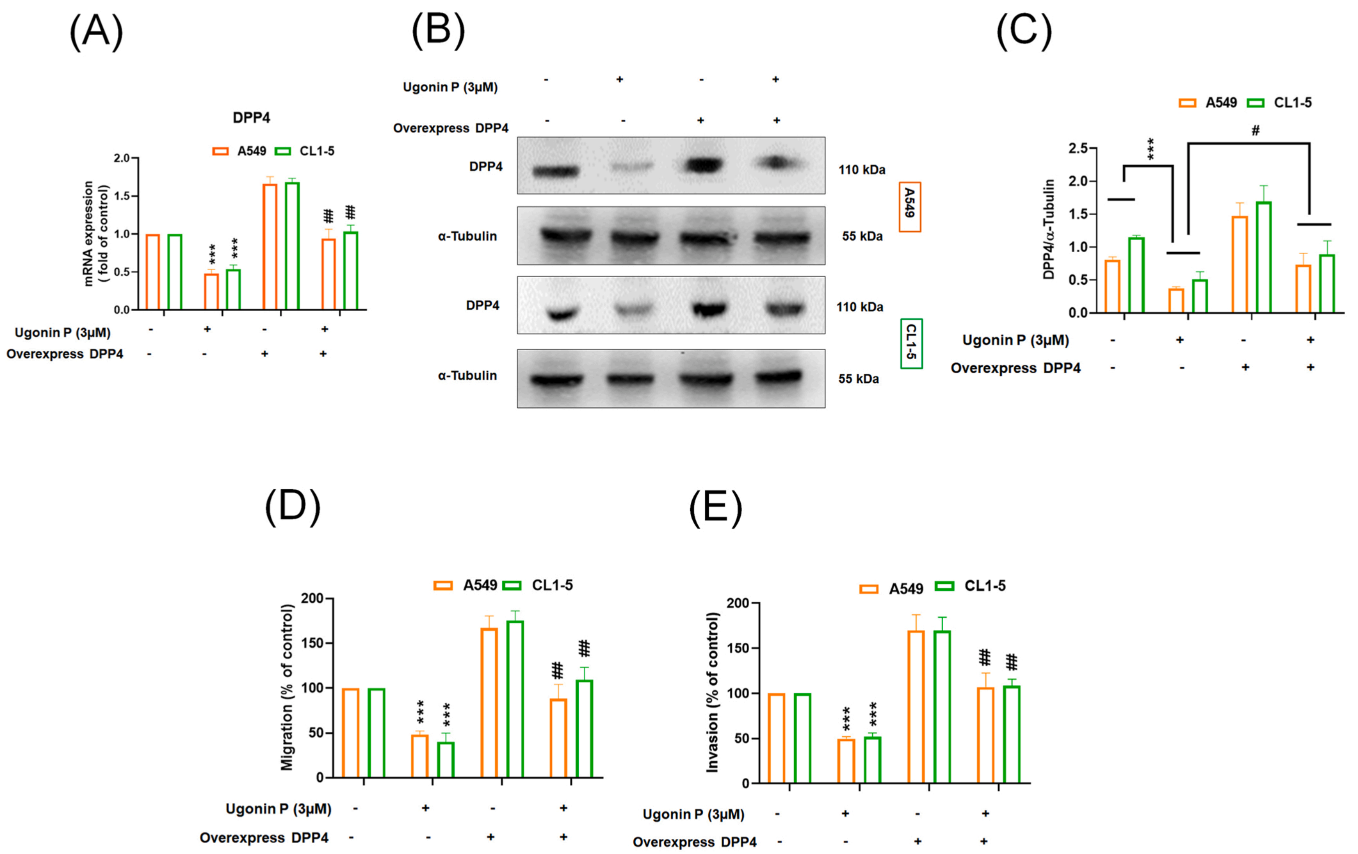

Fig. 5. Ugonin P inhibited DPP-4-dependent migration and invasion of lung cancer cells. A549 and CL1–5 cells transfected with DPP-4 plasmids and then incubated with ugonin P (3 μM) for 24 h. (A-C) qPCR and Western blot analysis of DPP-4 mRNA and protein expression, respectively (n = 3). Immunoblotting results was quantified by ImageJ software. α-Tubulin was used as loading control. (D&E) Transwell assay results indicating cell migration and invasion activity (n = 3). Results are expressed as the means ± SD. $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ vs. control group. $^{\#}p < 0.05$; $^{\#\#}p < 0.01$; $^{\#\#\#}p < 0.001$ vs. ugonin P-treated group.

DPP-4 synthesis, cell migration, and cell invasion (Fig. 8). Thus, it appears that ugonin P inhibits lung cancer motility and DDP-4 expression by upregulating the synthesis of miR-130b–5p.

4. Discussion

The International Agency of Research on Cancer Global Cancer Observatory provides data related to cancer rates (36 types) and mortality around the globe (185 countries) [37]. Lung cancer commonly metastasizes to the brain, bone, lymph nodes, and liver [38]. The highly intricate process of metastasis in lung cancer requires the participation of both the microenvironment surrounding the tumor as well as the stem cells from which the tumor develops [39]. Cancer cell metastasis to distant sites relies on the acquisition of migratory and invasive capabilities, which involves the induction of epithelial-mesenchymal transition [40]. Lung cancer motility is influenced by multiple cellular factors, including growth factors, cell adhesion molecules, chemokines, and pro-inflammatory cytokines [39].

Our analysis of markers from a human protease antibody array revealed that the expression of DPP-4 was lower than that of cancer-related protease protein markers. DDP-4 is a cell surface glycoprotein expressed in a variety of cell types, including epithelial and endothelial cells in the lung [32]. Depending on the cancer type and microenvironment, DPP-4 exhibits diverse expression levels and biological functions associated with cancer progression, including the modulation of cytokines and chemokines as well as interactions with the extracellular matrix [32]. DPP-4 has been identified in a variety of tumors, including ovarian cancer [41], colorectal cancer [42], pancreatic cancer [43], urothelial carcinoma [44], and lung cancer. Note that DPP-4 expression is strongly correlated with disease aggressiveness [32]. One previous study on various histological subtypes of lung cancer reported that most adenocarcinoma cases (93.1 %) were positive for DPP-4 expression [32]. Another study reported that DPP-4 expression was four times higher in lung cancer tissue than in normal tissue. In yet another study, inhibiting DPP-4 was shown to slow lung cancer growth in primary models of mouse and human lung cancer [45]. In the current study, DPP-4 expression levels were higher in a lung cancer tissue array than in normal lung cancer tissue samples. We also determined that DPP-4 expression levels were higher in the metastatic stage and strongly correlated with lung cancer survival rates.

For years, a variety of DPP-4 inhibitors (e.g., vildagliptin) have been in routine clinical use for the treatment of type II diabetes aimed at preventing atherosclerosis and myocardial injury, improving endothelial dysfunction, and lowering blood pressure [46]. Note however that this treatment has been associated with serious adverse side effects, including cardiovascular disorders [47]. Traditional Chinese medicine has been practiced for thousands of years and has been widely recognized as a rich source of pharmacological agents applicable to the development of novel cancer treatments [48,49]. Derivatives of *H. Zeylanica* have long been used for the treatment of inflammation, fever, pneumonia, and other disorders [50]. This is the first study to demonstrate that ugonin P (a flavonoid derived from *H. Zeylanica)* could be used to hinder the motility of lung cancer cells without cytotoxic effects. This study also demonstrates the efficacy of DPP-4 inhibition as a potential avenue for the development of anti-cancer therapies.

The RAF, MEK, and ERK signaling cascades have the ability to initiate the activation of multiple transcription factors associated with metastasis [51]. These signaling pathways play a crucial role in cancer progression by regulating various multicellular functions within the tumor microenvironment. They also promote cancer cell migration and

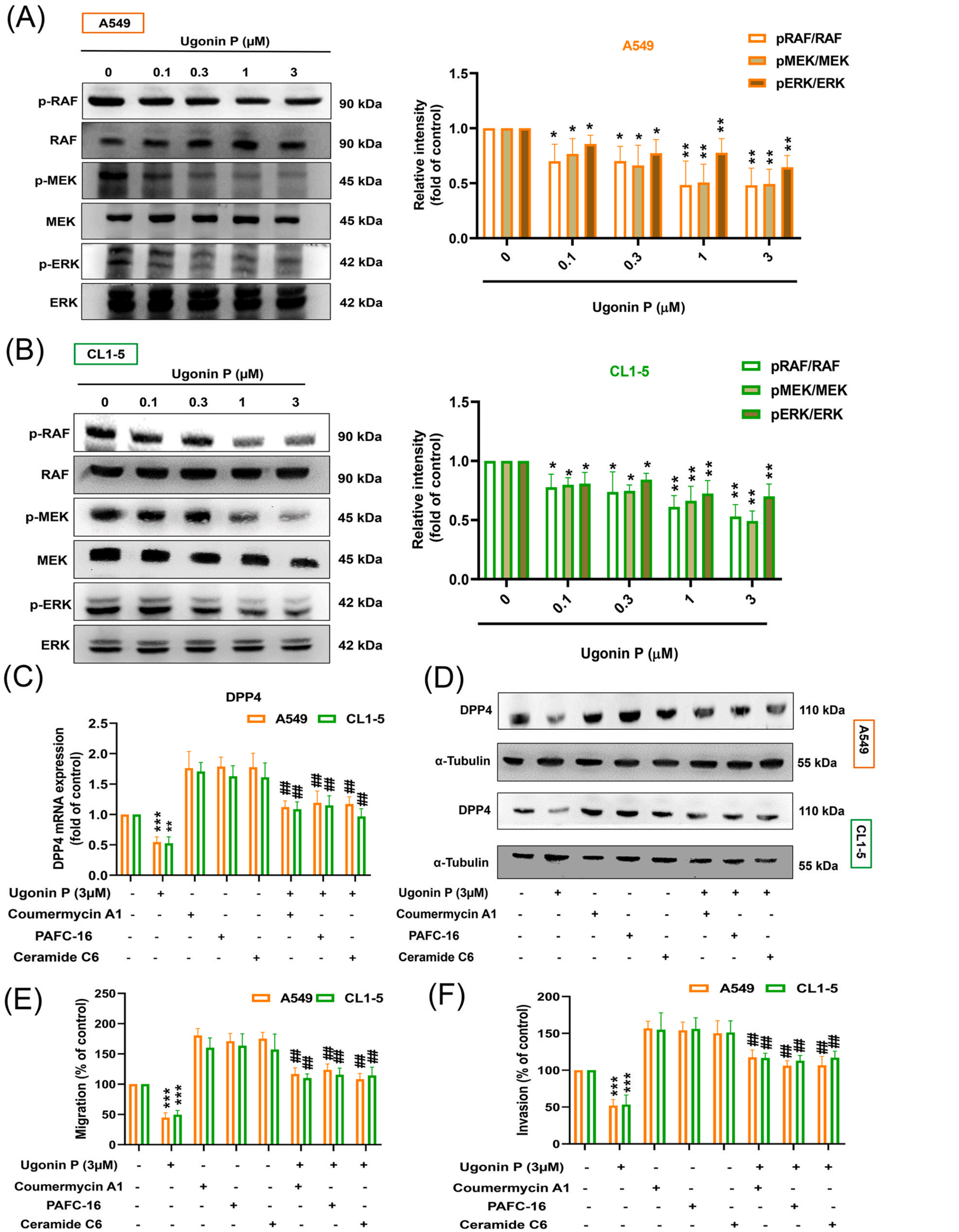

Fig. 6. RAF/MEK/ERK pathways were involved in ugonin P-induced inhibition of DPP-4 production and cell motility. Western blot analysis indicating RAF, MEK, and ERK phosphorylation in A549 (A) and CL1–5 (B) cell lines treated with ugonin P. Western blot was quantified by ImageJ software (n = 3). Cells were pre-treated with RAF/MEK/ERK activators for 30 min followed by ugonin P treatment (3 μM) for 24 h. (C-D) qPCR and Western blot analysis indicating DPP-4 mRNA and protein expression, respectively (n = 3). α-Tubulin was used as loading control. (E-F) Transwell assay results indicating cell migration and invasion capabilities (n = 3). Results are expressed as the means ± SD. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ vs. control group. $^{\#}$ $p < 0.05$; $^{\#\#}$ $p < 0.01$; $^{\#\#\#}$ $p < 0.001$ vs. ugonin P-treated group.

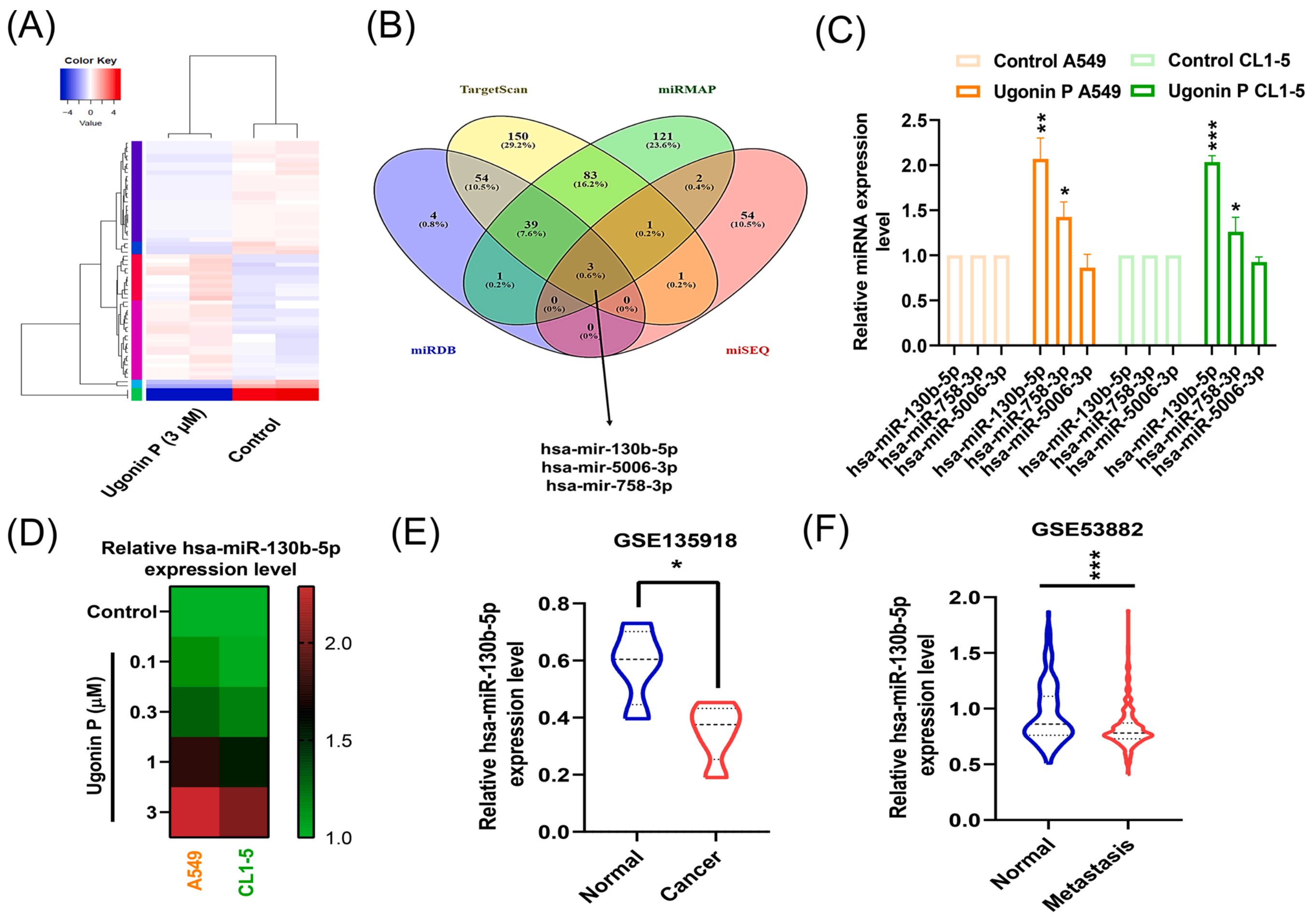

Fig. 7. Ugonin P inhibited DPP-4 expression by upregulating miR-130b-5p expression in lung cancer cells. (A&B) Differentially expressed heatmap indicating miRNA candidates that target DPP-4, based on the screening of data from a publicly available miRNA database merged with miRNA sequencing in A549 cells treated with or without ugonin P (3 μM). (C-D) Cells were incubated with ugonin P (3 μM) for 24 h, and miRNA expression levels were examined using qPCR (n = 3). (E-F) miRNA expression profiles of miR-130b-5p in lung cancer tissue and normal tissue based on the analysis of the GEO database. Results are expressed as the means ± SD. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ vs. control group.

contribute to the development of drug resistance [34,51]. The inhibition of RAF/MEK/ERK signaling pathways has been shown to stimulate immune-related activities aimed at eliminating cancer cells [52]. Our results in the current study indicated the involvement of RAF/MEK/ERK phosphorylation in the ugonin P-mediated inhibition of DPP-4 expression and cell motility. It appears that DPP-4 protein expression is mediated via the RAF/MEK/ERK pathway. Our results also suggest that the ugonin P-induced inhibition of DPP-4 expression is crucial to the motility of human lung cancer cells, thereby making the RAF/MEK/ERK pathway an attractive target for the development of therapeutics to combat lung cancer.

miRNAs are small non-coding RNA molecules that target mRNAs via base pairing in the 3’-untranslated regions of mRNA [53]. miRNA recognition leads to the cleaving or translational repression of the target mRNA [54], which plays a key role in the transcriptional and post-transcriptional regulation of gene expression [55]. Finally, miRNA plays a key role in cancer molecular biology, the identification of metastatic biomarkers, the regulation of angiogenesis, and the development of drug resistance [56]. One study reported that miR-137 expression levels were lower in lung cancer tissue than in normal tissue and that expression levels were correlated with disease progression [57]. They also reported that miR-137 suppressed the motility of lung cancer cells via epithelial-mesenchymal transition regulation. Another study reported that the ectopic expression of miR-16 suppressed cell proliferation and colony formation [58]. There is evidence to suggest that miR-130b-5p could serve as a biomarker indicating the development and/or progression of coronary artery disease [59]. In another study, miR-130b-5p over-expression was shown to inhibit SIRT4 expression and AMPK phosphorylation, which promotes the phosphorylation of Smad2 and Smad3 to accelerate the progression of liver fibrosis [60]. In the current study, we determined that miR-130b-5p expression levels were lower in lung cancer tissue than in normal tissue. We also determined that miR-130b-5p inversely regulates DPP-4 expression, thereby inhibiting the migration and invasion of lung cancer cells.

H. Zeylanica has reported to inhibit pro-inflammatory cytokines in macrophage cells [16]. Treatment of lung cancer cells with ugonin P also inhibited the expression of pro-inflammatory cytokines, including IL-1β, IL-6 and TNF-α expression (Supplementary Fig. S3). These results suggest ugonin P reduces the production of pro-inflammatory cytokines in the tumor microenvironment. Further examination is needed to determine whether ugonin P also regulates the immune system. A previous study indicated that ugonin J (50 mg/kg) reduces MCF-7 breast cancer growth in an orthotopic tumor model [61]. However, the isolation process for ugonin P is complicated and requires a long time to obtain a sufficient quantity of pure ugonin P. It is worth noting that a limitation of our study is the insufficient quantity of ugonin P available for in vivo studies. Further investigation is needed to determine whether ugonin P also inhibits lung cancer growth in vivo.

In conclusion, we determined that the metastasis of lung cancer cells depends on the migratory and invasive abilities facilitated by DPP-4 via

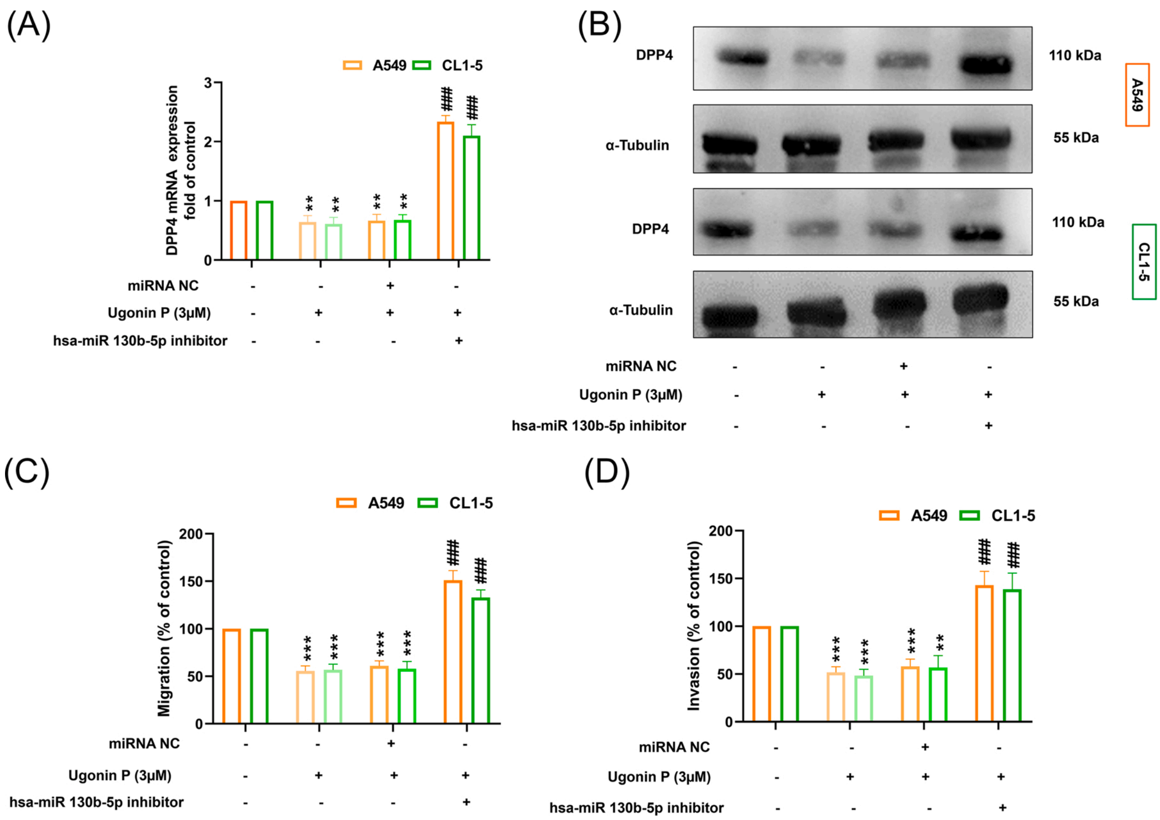

Fig. 8. miR-130b-5p is required for ugonin P-mediated inhibition of lung cancer cell migration and invasion. A549 and CL1-5 cells were transfected with a miR-130b-5p inhibitor, followed by 24 h of ugonin P treatment (3 μM). (A&B) qPCR and Western blot analysis indicating DPP-4 mRNA and protein expression levels, respectively (n = 3) α-Tubulin was used as loading control. (C&D) Transwell assays indicating cell migration and invasion capabilities (n = 3). Results are expressed as the means ± SD. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$ vs. control group. $^{\#}$ $p < 0.05$; $^{\#\#}$ $p < 0.01$; $^{\#\#\#}$ $p < 0.001$ vs. Ugonin P-treated group.

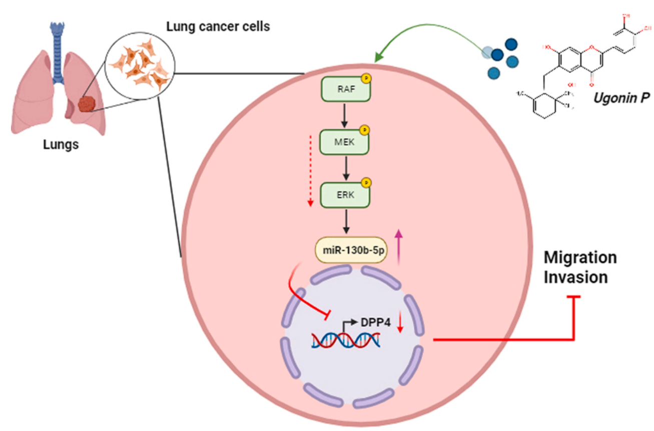

Fig. 9. Schema illustrating the effects of ugonin P on lung cancer cell motility. Ugonin P inhibited the DPP-4-dependent migration and invasion activity of lung cancer cells. Inhibition of the RAF/MEK/ERK pathway and upregulation of miR-130b-5p expression mediated the effects of ugonin P. (Schema was created using BioRender.com).

interactions with specific signaling pathways. The observed anti-motility effects of ugonin P in lung cancer cells involved down-regulating the RAF/MEK/ERK signaling cascades while simultaneously upregulating the expression of miR-130b-5p (Fig. 9). This is the first study to demonstrate the effects of uginon P (an extract of *H. Zeylanica*) in hindering the motility of cancer cells. This results provide compelling evidence that ugonin P has therapeutic potential for the development of therapeutic agents to treat lung cancer.

CRediT authorship contribution statement

Conceptualization, C.-Y.W., S.S.G., D.A. and C.-H.T.; methodology, S.S.G., and D.A.; software, C.-Y.W., S.S.G. and S.-Y.F.; validation, C.-Y. W., D.A. and P.-I.L.; formal analysis, C.-Y.W., C.-L.Huang. and C.L. Hsieh.; investigation, C.-Y.W. and S.S.G.; resources, C.-H.T. and C.-C.L.; data curation, S.S.G., and D.A.; writing-original draft preparation, D.A., and C.-H.T.; writing-review and editing, C.-T.H., and D.A.; visualization, C.-L.H.; supervision, C.L.Hsieh and C.-H.T.; project administration, C.L. Hsieh and C.-H.T.; funding acquisition, C.-C.L and C.-H.T. All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopha.2023.115483.

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